

STAT 301 In-class Activity No. 1

Consider the “wage” dataset posted on Canvas. The data consist of a random sample of 534 persons in US some years ago, with information on wages and other characteristics of the workers, including sex, number of years of education, years of work experience, occupational status, region of residence and union membership. Variable names are:

EDUCATION : Number of years of education .

SOUTH : Indicator variable for Southern Region

(1= Person lives in South , 0= Person lives elsewhere).

SEX : Indicator variable for sex (1= Female , 0= Male).

EXPERIENCE : Number of years of work experience .

UNION : Indicator variable for union membership (1= Union member , 0= Not union member)

WAGE : Wage (dollars per hour).

AGE : Age (years).

RACE : Race (1= Other , 2= Hispanic , 3= White).

OCCUPATION : Occupational category (1= Management , 2= Sales , 3= Clerical ,
4= Service , 5= Professional , 6= Other).

SECTOR : Sector (0= Other , 1= Manufacturing , 2= Construction).

MARR : Marital Status (0= Unmarried , 1= Married)

In-class activity

- (a) Plot WAGE versus EDUCATION. Is there a linear relationship between them?
- (b) Find the correlation coefficient of the two variables. Is it consistent with what you observe in (a)?
- (c) Fit a simple linear regression model, with WAGE as response and EDUCATION as a predictor. Is the result consistent with (a) and (b)?